# Frontend UI/UX Master Plan (Java Swing)

**Assignee:** Hanin (Lead UI/UX Designer)

**Project:** Hotel Booking System

**Backend Counterparts:** Adham & Sherif

**Logic Counterpart:** Alaa

**Deadline:** Phase 1 (All visual screens) must be completed within 4-5 days.

## Your Objective

As the Lead UI/UX Designer, you are the "architect of the user experience." Your primary mission is to design clean, modern, and user-friendly screens using **Java Swing**.

You do **NOT** need to worry about the database, SQL, or what happens when a button is clicked (Alaa will handle the button clicks and logic). Your job is to build the visual "shell" (JFrames, JPanels, buttons, and text fields) and make it look as professional as possible.

## Phase 1: Tooling & Setup (Work Smart)

Since we are on a tight 14-day sprint, **do not write GUI code entirely by hand**.

- **If you use IntelliJ IDEA,** use the built-in *GUI Designer*.
- **If you use Eclipse,** install the *WindowBuilder* plugin.
  These tools allow you to "drag and drop" buttons and text fields, which will save you days of work.
  *Note: Youssef will inject the "FlatLaf" theme later to make your Swing design look like a modern 2024 app, so focus purely on layout and spacing.*

## Phase 2: Required Screens (The Views)

You need to design the following JFrame / JPanel windows. To test them, put **Dummy Data** (fake text) in the tables and fields for now.

### 1. Login & Registration Screen

- **Components:** Email field (JTextField), Password field (JPasswordField), Login Button, and a clickable text/button to switch to the "Register" screen.
- **Layout Tip:** Use GridBagLayout or GridLayout to keep the text fields perfectly aligned in the center.

### 2. Admin Dashboard (Main Hub for Admins)

- **Components:** Use a JTabbedPane (tabs at the top or side) to separate tasks:
  - **Tab 1 (Manage Rooms):** A form to add a new room (room number, type combo box, and price) + "Add Room" button. Below it, a JTable displays all rooms.
  - **Tab 2 (View Bookings):** A large JTable showing all reservations made by customers.
  - **Tab 3 (Statistics):** Leave an empty JPanel here. Sherif and Youssef will later inject a pie chart into this space.

### 3. Customer Dashboard (Main Hub for Guests)

- **Components:** * **Top Panel:** Search and filter area (e.g., Max Price text field, Room Type combo box) + "Search" button.
  - **Center Panel:** A JTable showing only *available* rooms based on the search.
  - **Bottom Panel:** A "Book Selected Room" button.

## Phase 3: The "Handover" Rules (Crucial for Alaa)

Because you are designing the screens and **Alaa** is writing the logic (making the buttons actually work), Alaa needs to be able to access the buttons and text fields you create.

**Hanin's Golden Naming Conventions:**

Never leave a component named jButton1 or jTextField2. You must name them professionally:

- Text Fields: txtEmail, txtPassword, txtRoomPrice
- Buttons: btnLogin, btnAddRoom, btnBookRoom
- Tables: tblRooms, tblBookings
- Combo Boxes (Drop-downs): cmbRoomType

  **Visibility Rule:** Make sure the UI components that Alaa needs to interact with (like buttons and text fields) are set to public or have public getters in your GUI class.

## Phase 4: Swing UI/UX Best Practices

To ensure the professor gives us the "Bonus" mark for an outstanding GUI:

1. **Margins & Padding:** Don't let buttons and text fields touch the edges of the window. Use EmptyBorder on your JPanels (e.g., setBorder(new EmptyBorder(20, 20, 20, 20))) to give the UI room to "breathe."
2. **Standard Button Sizes:** Make sure buttons like "Login" and "Cancel" have the same width and height so the design looks symmetric.
3. **Fonts:** Use a clean, readable font (e.g., *Segoe UI*, size 14) for all components instead of the default Swing font.

**Next Steps for You:**

Open your IDE, sketch out the screens on a piece of paper, and start dragging and dropping! Let Alaa know as soon as the LoginScreen.java is visually ready so she can start adding the click events. Good luck!